

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester IV Examination, 2023-24

Botany**Paper: BOB-403A — Ancillary Botany-II**

Time: Three Hours

Maximum Marks: 70

Instructions: Answer five questions including Question 1 which is compulsory. The figures in the right-hand margin indicate marks.

Question 1. Fill in the blanks or answer the following questions (in not more than 50 words each): [10 × 1 = 10]

- (a) What is substrate-level phosphorylation?
- (b) What is proto-cooperation?
- (c) Define apical dominance.
- (d) What are hydrophytes?
- (e) Gibberellins were first recognised from the fungus _____.
- (f) What are the main constituents of *Tinospora cordifolia*?
- (g) Define Endemic disease.
- (h) What is the role of microconsumers in the ecosystem?
- (i) The disease Citrus canker is caused by _____.
- (j) Why are nutrient cycles also referred to as biogeochemical cycles?

Question 2. What are plant growth hormones? Describe the various roles of ethylene in detail in plants. [15]

Question 3. What are biotic interactions in the ecosystem? Describe different kinds of possible negative interactions among species. [15]

Question 4. Discuss the structural components and function of the ecosystem in detail. What is the importance of the food chain in the ecosystem? [15]

Question 5. Discuss how nutrient cycling takes place in the ecosystem. Explain the steps of the carbon cycle in the atmosphere. [15]

Question 6. Write short notes on any two of the following: [2 × 7.5 = 15]

- (a) Food pyramid
- (b) Disease symptoms and mode of spread and transmission of Papaya mosaic virus

(c) Sulphur cycle

Question 7. Write short notes on any two of the following: [2 × 7.5 = 15]

- (a) Medicinal uses and chemical constituents of *Withania somnifera*
- (b) Mutualism with suitable examples
- (c) Plant responses to changes in temperature

Question 8. Answer/write short notes on any three of the following: [3 × 5 = 15]

- (a) What is photoperiodism? Classify plants based on light period requirements.
- (b) Phloem loading
- (c) Structure of phytochrome
- (d) Electron transport system in mitochondria

Question 9. Define plant tissue culture. Describe in detail the various techniques of plant tissue culture along with their applications. [15]

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